

ANSWER

Q1. What is Linux, and how is it different from Windows?

Linux is an open-source, Unix-like OS kernel.

- Free & customizable.
 - Windows is proprietary, GUI-focused; Linux is CLI-driven but also supports GUIs.
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Q2. How do you check the current Linux version and kernel version?

```
cat /etc/os-release # OS version
```

```
uname -r           # Kernel version
```

Q3. What is the difference between absolute and relative paths?

- **Absolute path:** starts from / (root). Example: /home/user/file.txt
 - **Relative path:** relative to the current directory. Example: ./file.txt
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Q4. Difference between soft link and hard link?

- **Soft link (symlink):** points to the file's path, can cross partitions.
- **Hard link:** points to the file's inode, cannot cross partitions.

```
ln file.txt hardlink.txt # Hard link
```

```
ln -s file.txt softlink.txt # Soft link
```

Q5. How do you check disk usage and free space?

```
df -h # disk usage by filesystem
```

```
du -sh * # size of files/folders in current dir
```

Q6. How do you check memory and CPU usage?

```
free -h # memory usage
```

```
top      # CPU + processes
```

htop # (if installed, interactive view)

Q7. What is the difference between /etc/passwd and /etc/shadow?

- /etc/passwd → stores usernames, UID, GID, home, shell.
 - /etc/shadow → stores **encrypted passwords** & expiry info (restricted access).
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Q8. How do you create a new user and assign a password?

sudo useradd newuser

sudo passwd newuser

Q9. How do you schedule jobs?

- **cron** → recurring jobs

crontab -e

Example: Run script.sh every day at 2 AM

0 2 * * * /path/script.sh

- **at** → one-time jobs

at 5pm

at> /path/script.sh

Q10. How do you change permissions and ownership?

chmod 755 file.sh # permissions

chown user:group file # ownership

Q11. Difference between find and locate?

- **find** → real-time search

find /home -name file.txt

- **locate** → uses prebuilt database (updatedb) → faster but not real-time.

locate file.txt

Q12. Where are logs stored in Linux?

Mostly under /var/log/

Examples:

- `/var/log/messages` → system logs
 - `/var/log/secure` → authentication
 - `journalctl -xe` → systemd logs
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Q13. Difference between TCP and UDP?

- **TCP** → Connection-oriented, reliable, slower (e.g., HTTP, SSH).
 - **UDP** → Connectionless, fast, no guarantee (e.g., DNS, streaming).
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Q14. What is SELinux?

Security-Enhanced Linux → provides **mandatory access control (MAC)**.

- Modes: enforcing, permissive, disabled.

`getenforce` # check mode

`setenforce 0` # set permissive

Q15. What is LVM, and how do you create one?

Logical Volume Manager = flexible disk management.

`pvcreate /dev/sdb`

`vgcreate my_vg /dev/sdb`

`lvcreate -L 5G -n my_lv my_vg`

`mkfs.ext4 /dev/my_vg/my_lv`

`mount /dev/my_vg/my_lv /mnt`

Q16. What is RAID? Difference between RAID 0, 1, 5, 10?

- **RAID 0** → Striping, performance, no redundancy.
 - **RAID 1** → Mirroring, redundancy.
 - **RAID 5** → Striping + parity, needs ≥ 3 disks.
 - **RAID 10** → Mirroring + striping, performance + redundancy.
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Q17. What is the difference between SAN, NAS, and DAS?

- **DAS** → Direct Attached Storage (local disk).
- **NAS** → Network Attached Storage (file-level).
- **SAN** → Storage Area Network (block-level, e.g., iSCSI, Fibre Channel).

Q18. How do you mount an NFS share?

`sudo mount -t nfs server:/share /mnt`

To make permanent: add entry in `/etc/fstab`.

Q19. What is iSCSI? How do you configure it?

iSCSI = block-level storage over IP.

- Configure target on storage server.
- Discover & login from initiator:

`iscsiadm -m discovery -t sendtargets -p <target-ip>`

`iscsiadm -m node -T <IQN> -p <target-ip> --login`

Q20. Difference between init and systemd?

- **init** → Older, sequential boot.
 - **systemd** → Modern, parallel boot, uses `systemctl`.
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Q21. Server is slow, how do you troubleshoot?

- Check CPU/memory: `top`, `free -h`
 - Check disk usage: `df -h`, `iostat`
 - Check logs: `dmesg`, `journalctl`
 - Check running processes: `ps aux`
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Q22. A user cannot log in — what do you check?

- Account locked? → `passwd -S user`
 - Expired password? → `chage -l user`
 - Logs in `/var/log/secure`
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Q23. Disk full, how do you free space?

`df -h`

`du -sh /* | sort -h`

Delete old logs (/var/log/), remove temp files.

Q24. A service is not starting — steps?

systemctl status servicename

journalctl -xe

Check config files & logs.

Q25. Network is unreachable, how do you debug?

ping 8.8.8.8 # connectivity

ip addr # check IP

ip route # check gateway

systemctl restart network

Q1. What is Linux, and how is it different from UNIX?

Linux is an open-source OS kernel, UNIX is proprietary. Linux is community-driven, runs on multiple platforms, while UNIX is vendor-specific.

Q2. Explain the Linux directory structure.

Root / contains:

- /bin (user binaries), /sbin (system binaries), /etc (configs), /var (logs), /home (user dirs), /tmp (temporary).

Q3. Difference between relative and absolute paths?

- Absolute → starts from / (e.g., /home/user/file).
- Relative → from current dir (e.g., ./file).

Q4. What are hard links and soft links?

- Hard link → points to inode, survives file deletion.
- Soft link → shortcut, breaks if target deleted.

Q5. Difference between Linux and Windows file systems?

- Linux supports ext4, xfs, btrfs (case-sensitive).
- Windows uses NTFS/FAT (case-insensitive).

Q6. Explain file permissions in Linux.

- r (read), w (write), x (execute).
- Example: -rw-r--r-- → owner read/write, group read, others read.

Q7. Difference between chmod, chown, and chgrp?

- chmod → change file permissions.
- chown → change file owner.
- chgrp → change group.

Q8. How do you check system uptime?

uptime

Q9. How do you find the current logged-in users?

who

w

Q10. Difference between /bin, /sbin, /usr/bin, /usr/sbin?

- /bin → essential user commands.
 - /sbin → essential system commands.
 - /usr/bin → additional user programs.
 - /usr/sbin → non-essential system commands.
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Q11. Explain the Linux boot process.

1. BIOS/UEFI loads bootloader (GRUB).
2. Kernel loads.
3. init/systemd starts services.
4. Login prompt provided.

Q12. What are runlevels / systemd targets?

- Runlevels: 0 (halt), 1 (single-user), 3 (multi-user), 5 (GUI), 6 (reboot).
- systemd uses targets (e.g., graphical.target, multi-user.target).

Q13. How do you check memory and CPU usage?

free -h

top

htop

Q14. Difference between process and thread?

- Process → independent program with its memory.
- Thread → lightweight, shares memory with parent process.

Q15. Difference between ps, top, and htop?

- ps → snapshot of processes.
- top → real-time process view.
- htop → interactive version of top.

Q16. How do you schedule tasks in Linux? Difference between cron and at?

- cron → repetitive tasks.
- at → one-time jobs.

Q17. Difference between /etc/passwd and /etc/shadow?

- /etc/passwd → user info (UID, GID, shell).
- /etc/shadow → encrypted passwords + expiry.

Q18. How do you create a new user and assign groups?

useradd -G developers user1

passwd user1

Q19. Explain sticky bit, SUID, and SGID.

- **Sticky bit:** only owner can delete (used in /tmp).
- **SUID:** run with owner's privileges.
- **SGID:** run with group's privileges.

Q20. How do you check which processes are using a file or port?

lsof /path/file

lsof -i :80

Q21. Difference between TCP and UDP?

- TCP: reliable, connection-oriented (SSH, HTTP).
- UDP: fast, no guarantee (DNS, streaming).

Q22. How do you troubleshoot network connectivity in Linux?

- ping, ip addr, ip route, ss -tulnp, tcpdump.

Q23. Difference between SAN, NAS, DAS?

- DAS → directly attached storage.
- NAS → network file-level storage.
- SAN → block-level storage over network.

Q24. What is LVM? How do you create and extend a logical volume?

👉 LVM = Logical Volume Manager.

`pvccreate /dev/sdb`

`vgcreate vg1 /dev/sdb`

`lvcreate -L 5G -n lv1 vg1`

`mkfs.ext4 /dev/vg1/lv1`

`mount /dev/vg1/lv1 /mnt`

Extend:

`lvextend -L +2G /dev/vg1/lv1`

`resize2fs /dev/vg1/lv1`

Q25. Difference between ext3, ext4, and xfs?

- ext3 → journaling.
- ext4 → faster, bigger volumes.
- xfs → best for large files.

Q26. What is RAID? Explain RAID levels.

- RAID 0 → striping (performance).
- RAID 1 → mirroring (redundancy).
- RAID 5 → striping + parity.
- RAID 10 → mirroring + striping.

Q27. What is SELinux/AppArmor?

- Security frameworks enforcing access control.
- Modes: enforcing, permissive, disabled.

Q28. What is iSCSI, and where is it used?

- iSCSI = block storage over IP.
- Used in SANs, virtualization, central storage.

Q29. Difference between init and systemd?

- init → old, sequential.
- systemd → new, parallel, manages services better.

Q30. How do you mount NFS and CIFS shares?

`mount -t nfs server:/share /mnt`

`mount -t cifs //server/share /mnt -o user=username`

Q31. A user cannot log in — how do you troubleshoot?

- Check if account locked: `passwd -S user`
- Expired? `chage -l user`
- Logs: `/var/log/secure`

Q32. A server is running slow — what steps?

- Check CPU/memory: `top`, `free -h`
- Check disk: `iostat`, `df -h`
- Check logs: `dmesg`, `journalctl`

Q33. Disk space full — how do you clear?

`df -h`

`du -sh /*`

Delete old logs/temp files.

Q34. A service is not starting — how do you debug?

`systemctl status service`

`journalctl -xe`

Q35. Network unreachable — how to debug?

- ping, ip addr, ip route, check gateway.

Q36. Server not booting — how do you fix?

- Boot in rescue mode.
- Check `/boot/grub` configs.
- Check logs: `/var/log/boot.log`.

Q37. Logs filling disk — how to fix?

- Rotate logs (`logrotate`).
- Clear old logs (`/var/log/`).

Q38. High I/O wait — how to check?

`iostat -xz 1`

Check heavy processes with `iostat`.

Q39. How do you recover a deleted file?

- If open, use `/proc/<PID>/fd`.
- Otherwise, restore from backup.

Q40. How do you secure SSH on Linux?

- Disable root login (`/etc/ssh/sshd_config`).

- Use SSH keys instead of passwords.
 - Change default port.
 - Use firewall/Fail2Ban.
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